

AI-Driven Drug Repurposing

Presentation & Viva Preparation Notes

Project pipeline: Existing drug → Molecular targets → Open Targets disease associations → Transparent AI ranking → PubMed literature validation → Candidate interpretation

1. Project in One Minute

Drug repurposing means finding a new therapeutic use for an existing drug. Instead of starting drug discovery from zero, the project uses existing drug–target and disease–association evidence to prioritize diseases that may be worth investigating.

The system is designed as an **AI-assisted hypothesis-generation tool**. It does not claim that a drug is clinically effective for a newly ranked disease.

In this project, Open Targets provides target–disease association evidence. A transparent scoring formula combines the strongest association, average association, and the number of supporting targets. PubMed is then used as a literature-validation layer.

2. Why This Project?

Traditional drug discovery is expensive and time-consuming. Drug repurposing can reduce the amount of early discovery work because the drug may already have known pharmacological and safety information.

AI and biomedical databases can help researchers search large numbers of drug–target–disease relationships and prioritize hypotheses for further experimental investigation.

3. System Architecture

Step	What happens
1. Drug input	User enters a drug such as Aspirin or Metformin.
2. Drug identification	The drug is mapped to its ChEMBL/Open Targets identifier.
3. Target retrieval	Mechanisms of action provide molecular targets.
4. Disease associations	Open Targets supplies target–disease association scores.
5. AI ranking	A transparent heuristic produces an AI Repurposing Score.
6. Literature validation	Drug–disease combinations are searched in PubMed.
7. Dashboard	Results are shown as tables, score graphs and a drug–target–disease network.

4. AI Repurposing Score

The current score is a **transparent heuristic**, not a clinically validated machine-learning model:

AI Repurposing Score = $0.60 \times \text{Max Association Score} + 0.30 \times \text{Mean Association Score} + 0.10 \times \text{normalized Supporting Targets}$

The maximum association captures the strongest evidence from a target. The mean score summarizes evidence across supporting targets. Supporting-target count rewards diseases supported by multiple drug targets, with the contribution capped after five targets.

A high score means **stronger computational support within this ranking system**; it does not mean a high probability of clinical success.

5. Example: Aspirin

The current Open Targets analysis identifies **CHEMBL25** for Aspirin and returns two molecular targets: **PTGS1** and **PTGS2**.

Rank	Disease	AI Score
1	Rheumatoid arthritis	0.608498
2	Gout	0.604963
3	Migraine disorder	0.604101
4	Hemorrhage	0.599086
5	Osteoarthritis	0.598830
6	Headache	0.594823
7	Ulcerative colitis	0.591608
8	Dysmenorrhea	0.590469
9	Pain	0.588253
10	Fever	0.587897

Important: rheumatoid arthritis, gout, migraine, pain, fever and other entries can represent established relationships. The ranking is therefore a prioritization output, not a claim of discovering a previously unknown indication.

6. PubMed Validation

PubMed is used to check whether literature exists for the combination **drug + disease**. This adds an evidence layer after computational ranking.

A PubMed hit indicates that relevant literature may exist. It does **not** by itself prove that the drug works for that disease. Articles can include reviews, case reports, mechanistic studies, observational studies or other evidence types.

7. How to Explain the Dashboard

When demonstrating the app, enter **Aspirin**. Explain the result in this order:

- **Drug:** Aspirin is identified through its database identifier.
- **Targets:** PTGS1 and PTGS2 are retrieved from its mechanisms of action.
- **Disease associations:** Open Targets provides association scores for diseases linked to those targets.
- **AI ranking:** the transparent formula prioritizes the disease candidates.
- **Graphs:** the dashboard makes the ranking and target support easier to interpret.
- **PubMed:** literature presence is checked for the top candidates.

8. Viva Questions — Most Important

Q. What is drug repurposing?

A. Finding a new therapeutic use for an existing drug.

Q. Why use AI?

A. AI-assisted methods can prioritize large numbers of possible drug–disease relationships and reduce manual screening.

Q. What database did you use?

A. Open Targets for drug/target and target–disease evidence; PubMed for literature validation.

Q. What is ChEMBL?

A. A major bioactivity and drug-related database used to identify compounds and their biological interactions.

Q. What is Open Targets?

A. A platform that integrates evidence linking targets with diseases to support drug-target discovery and prioritization.

Q. What are PTGS1 and PTGS2?

A. They encode cyclooxygenase enzymes involved in prostaglandin synthesis and are molecular targets of Aspirin.

Q. Is your AI model clinically validated?

A. No. It is a transparent computational ranking/heuristic intended for hypothesis generation.

Q. Does a high AI score mean the drug will treat the disease?

A. No. It means the disease has stronger support according to the selected computational scoring system.

Q. Why use PubMed?

A. To check whether published scientific literature exists for a drug–disease pair.

Q. Does a PubMed hit prove efficacy?

A. No. Literature presence does not establish efficacy, causality, safety or regulatory approval.

Q. What is the biggest limitation?

A. Database evidence and literature co-occurrence can contain bias and do not replace experimental or clinical validation.

Q. What is the future scope?

A. Add more evidence types, validated machine-learning models, stronger literature classification, clinical-trial evidence, safety data and experimental validation.

9. 30-Second Presentation Script

"My project is an AI-assisted drug repurposing analyzer. The aim is to prioritize potential new disease applications for existing drugs. The system identifies the drug and its molecular targets using biomedical database information, retrieves target–disease associations from Open Targets, and calculates a transparent AI Repurposing Score using maximum association, mean association and supporting-target information. The top candidates are then checked against PubMed literature. The final dashboard presents the results using rankings, graphs and a drug–target–disease network. Importantly, the system generates research hypotheses and does not claim clinical efficacy, safety or approval."

10. Strong Closing Line

"The value of this system is not replacing experimental research; it is helping researchers decide **which hypotheses deserve attention first.**"

11. Scientific Safety Statement

Computational predictions and PubMed co-occurrence do not establish drug efficacy, safety, causality, clinical benefit, or regulatory approval. All candidates require appropriate experimental and clinical validation.